



RITIS Filter API Reference 5.2.57 OAS3

<https://filter.ritis.org/v3/api-docs>

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1. Overview

The **RITIS Filter API** allows users to query various types of data from the RITIS databases, e.g. traffic incidents, dynamic message signs, weather.

This document is a work in progress and subject to change at any time (see [change log](#)).

If you have additional questions, please email us at support@ritis.org.

1.1 Authentication

RITIS Filter is currently restricted to authorized users. You are logged in as [gur002@henrico.gov](#). Your API Key is **2b27cbe8f5db4d0997e9c0d405b0dec1**.

To authenticate GET requests, add a parameter **api-key** with your API key as the value:

https://filter.ritis.org/ritisfilter/event/?api-key=your_api_key

To authenticate POST requests, add an HTTP header **X-RITIS-Filter-API-Key** with your API key as the value:

POST /incident HTTP/1.1
Host: <https://filter.ritis.org>
X-RITIS-Filter-API-Key: your_api_key

If you need an additional API key for your system, please email us at support@ritis.org.

Please limit the request rate to **10 times per minute per API key**.

1.2 Sending filter requests

1.2.1 GET

Filter parameters are passed as GET parameters in the URL. For example, you can use the road filter to get all active events on I-495 as reported by the Maryland DOT:

https://filter.ritis.org/ritisfilter/event/?api-key=your_api_key&system=mdot_chart&road=I-495

The [/event](#) endpoint returns event data. For other data types, see [Endpoints](#).

For a complete list of GET parameters and detailed information regarding each one, see [Filters](#).

1.2.2 POST

Filter parameters are encoded in an XML document, then sent in the HTTP message body of a POST request. For example, the same request for events on I-495 in XML form:

```
<event-filter xmlns="http://www.ritis.org/schema/filter">
  <systems>
    <system>mdot_chart</system>
  </systems>
  <location-filters>
    <road-filters>
      <road>I-495</road>
    </road-filters>
  </location-filters>
</event-filter>
```

can be sent to:

<https://filter.ritis.org/ritisfilter/event/>

The root element of the XML document indicates the type of data requested: **<event-filter>** for events, **<dms-filter>** for DMS data, etc.

1.2.3 Input schema

XML documents in POST requests conform to an [input schema](#). Validating your requests against this schema helps you debug bad request errors.

1.2.4 How filters are applied

Multiple different filters are joined with AND, such as road is 'I-95' AND state is 'MD'.

Multiple filters of the same type are joined with OR, such as road is 'I-95' OR road is 'I-495', except for geospatial filters (box filter, radius filter).

Multiple geospatial filters of the same type are joined with AND: two box filters that do not overlap will return no results.

1.3 Output formats

The API can return data in multiple formats. You can specify which using [output parameters](#). Detailed information regarding the output of each endpoint can be found in the [Endpoints](#) section. Within the endpoint's dropdown, under the Responses header, next to Example Values, click the Schemas link to see a detailed schema of the response with descriptions for each tag.

1.3.1 ATIS/TMDD Schema

The ATIS/TMDD schema is an official standardized schema for traffic events, DMS & detector data. The CATT Lab includes a few extensions for information that is missing but deemed critical.

You can find the complete ATIS & TMDD schema [here](#).

Lanes are arranged like this:

S S S S N N N N
0 1 2 3 4 5 6 7

RF will return the lanes in the order:

3 2 1 0 4 5 6 7

Thus the lanes are ordered from left to right in the order of travel *per* direction, regardless of the number of directions.

Lanes from the same direction are always grouped together, and directions are ordered, from left to right: south before north, west before east, and inner loop before outer loop.

1.3.2 CATT Lab Schema (rf-schema-xml)

The CATT Lab developed an unofficial schema for traffic events, DMS, & detector data based on our experiences working with the data. In general, this schema can provide more data. It can also express how an event changed over time for better analysis without complicated querying.

You can find the complete CATT Lab schema [here](#).

1.3.3 CATT Lab JSON Schema (rf-json)

The structure of this schema matches the regular CATT Lab Schema (rf-schema-xml) almost exactly. Some elements have been restructured to follow JSON standards. For example, "event" tags in rf-schema-xml are enclosed in another tag called "events". In the JSON output, the "event" tags are removed and the "events" tag is turned into a JSON array.

This schema is under development and **not finalized**.

2. Available agencies

These are all of the available agencies (system IDs) by data type.

Use the [Agency API](#) to learn more.

Crash
Detector
Device

Dms
Evacuation
Event
Other
Transit
Weather

3. Filters

3.1 General Filters

Active Only Filter	Indicate whether to include expired/closed/inactive results
Data Type	Specify the data type
Mode Filter	Limit results to one or more modes
Scenario Code Filter	Limit evacuation results to the specified scenarios
System (Agency) Filter	Limit results to one or more systems

3.2 crash-filters

Collision Type Filter	Limit results to those with a certain collision type
Light Condition Filter	Limit results to those with a certain light condition
Road Surface Condition Filter	Limit results to those with a certain road surface condition
Weather Condition Filter	Limit results to those with a certain weather condition

3.3 id-filters

Associated ID Filter	Return the item with the specified associated ID
ID Filter	Return the item with the specified ID

3.4 lane-filters

Closed Lane Filter	Limit results to events with the specified number or percentage of closed lanes
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3.5 location-filters

Box Filter	Limit results to within (or not within) a box
City Filter	Filter results by city
Country Filter	Filter results by country
County Filter	Filter results by county
Lat Lon Box Filter	Limit results to within (or not within) a box given southwest and northeast latlon corner points
Radius Filter	Limit results to within (or not within) a distance from a center point
State Filter	Filter results by state
Unmapped Filter	Indicate whether to return results with no location data (latitude/longitude)

3.6 location-filters/road-filters

Mile Marker Filter	Limit results to those within a range of mile markers on a given road
Road Class Filter	Limit to results that are part of the FRC (Functional Road Class)
Road Name Filter	Filter results by roads

3.7 location-filters/tmc-filter

TMC Filter	Limit results to the specified TMC area
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3.8 location-filters/ts-filter

TS Segment ID Filter	Limit results to the specified TS segment IDs
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3.9 location-filters/xd-filter

XD Segment ID Filter

Limit results to the specified XD segment IDs

3.10 output-parameters**Exclude Geometry**

Indicates whether geometry data should be excluded in the response

Group By

Aggregation grouping by a particular field. This is only usable with the Aggregation API. Uses in other APIs will return a 400 error.

Order

Sort results by the specified property

Output Format

Specify the output format

Response Limit

Limit the number of results

Skip Caches

Indicate whether to skip the cache and force a database query (use sparingly)

XSL

Use a previously uploaded XSL file to transform the XML result

3.11 request-headers**API Key**

Specify the API key

3.12 time-filters**Active During Filter**

Limit to results active/open at some point during the specified times

Cleared Time Filter

Limit to results cleared between the specified times. Precisions greater than a millisecond are not supported.

Confirmed Time Filter

Limit to results confirmed between the specified times

Created Time Filter

Limit to results created between the specified times

End Time Filter

Limit to results which ended between the specified times

Planned During Filter

Limit to events planned during the specified times

Start Time Filter

Limit to results which started between the specified times

Updated Time Filter	Limit to results updated between the specified times
Updated Within Filter	Limit to results updated within the specified number of seconds before request time.

3.13 time-filters/dow-filter

Day of Week Filter	Limit to results start in the specified days of week
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3.14 time-filters/hour-of-day-filter

Hour of Day Filter	Limit results to those within certain hours of the day
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3.15 type-filters

Agency Event Subtype Filter	Limit to traffic events of the specified agency-specific subtype
Agency Event Type Filter	Limit to traffic events of the specified agency-specific type
Event Subtype Filter	Limit to traffic events of the specified subtype
Event Type Filter	Limit to traffic events of the specified type
Every Day Counts Incident Type Filter	Limit to traffic events of the specified EDC incident type

4. Endpoints

Servers

https://filter.ritis.org - RITIS Filter Server

4.1 Documentation



GET /reference Complete documentation



GET /reference/public Public documentation



4.2 Download CCTV images



GET /ritisfilter/media/cctv/{id} Download CCTV images for a given device



4.3 Download schemas for RITIS Filter



GET /reference/schema/inflexible-agency.xsd Old filter format for agency users



4.4 Explore, view, and download ZIPs of RITIS Filter schema files



GET /reference/schema/explorer Explorer of the RITIS Filter schema files



4.5 Download/upload media items associated with an event



GET /ritisfilter/media/download Download media item with a given ID



4.6 GeoRSS feeds



GET /rss/event Query for events, sorted most recent first by default



4.7 List agencies providing data for RITIS



GET	/agencies	List all unique agencies	✓
GET	/agencies/type	List all agencies grouped by types of data provided	✓
GET	/agencies/type/{type}	List all agencies providing a data type	✓
GET	/agencies/type/{type}/{subtype}	List all agencies providing a data type and subtype	✓

4.8 Query for MDOT CHART WZDx events

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GET	/wzdx_v4.1/mdot.geojson	GET request for MDOT CHART WZDx events	✓
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4.9 Query for VDOT WZDx events

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GET	/wzdx_v4.0/vdot.json	GET request for VDOT WZDx events	✓
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4.10 Query historical DMS data

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GET	/ritisfilter/dms/historical_dms	GET request for historical DMS messages for a given list of DMS	✓
POST	/ritisfilter/dms/historical_dms	POST request for historical DMS messages for a given list of DMS	✓

4.11 Query historical ridership data

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GET	/ritisfilter/transit/historical_ridership	GET request for historical ridership statuses for a given list of agency ids	✓
POST	/ritisfilter/transit/historical_ridership	POST request for historical ridership statuses for a given list of agency ids	✓

4.12 Query historical detector data

GET**/ritisfilter/detector/historical_detector**GET request for historical detector data for a given list of detector IDs **POST****/ritisfilter/detector/historical_detector**POST request for historical detector data for a given list of detector IDs 

4.13 Query historical weather data

GET**/ritisfilter/weather/historical_weather**GET request for historical weather data for a given list of site IDs **POST****/ritisfilter/weather/historical_weather**POST request for historical weather data for a given list of site IDs 

4.14 Query the RITIS Filter API

GET**/ritisfilter/crash**Query using GET requests. Filters are passed as GET parameters in the URL. **POST****/ritisfilter/crash**Query using POST requests. Filters are defined as an XML document in the request body. **GET****/ritisfilter/detector**Query using GET requests. Filters are passed as GET parameters in the URL. **POST****/ritisfilter/detector**Query using POST requests. Filters are defined as an XML document in the request body. **GET****/ritisfilter/device**Query using GET requests. Filters are passed as GET parameters in the URL. **POST****/ritisfilter/device**Query using POST requests. Filters are defined as an XML document in the request body. **GET****/ritisfilter/dms**Query using GET requests. Filters are passed as GET parameters in the URL. **POST****/ritisfilter/dms**Query using POST requests. Filters are defined as an XML document in the request body. **GET****/ritisfilter/event**Query using GET requests. Filters are passed as GET parameters in the URL. 

POST	/ritisfilter/event	Query using POST requests. Filters are defined as an XML document in the request body.	✓
GET	/ritisfilter/incident	Query using GET requests. Filters are passed as GET parameters in the URL.	✓
POST	/ritisfilter/incident	Query using POST requests. Filters are defined as an XML document in the request body.	✓
GET	/ritisfilter/incidentevent	Query using GET requests. Filters are passed as GET parameters in the URL.	✓
POST	/ritisfilter/incidentevent	Query using POST requests. Filters are defined as an XML document in the request body.	✓
GET	/ritisfilter/other	Query using GET requests. Filters are passed as GET parameters in the URL.	✓
POST	/ritisfilter/other	Query using POST requests. Filters are defined as an XML document in the request body.	✓
GET	/ritisfilter/transit	Query using GET requests. Filters are passed as GET parameters in the URL.	✓
POST	/ritisfilter/transit	Query using POST requests. Filters are defined as an XML document in the request body.	✓
GET	/ritisfilter/weather	Query using GET requests. Filters are passed as GET parameters in the URL.	✓
POST	/ritisfilter/weather	Query using POST requests. Filters are defined as an XML document in the request body.	✓
GET	/ritisfilter/transit/agency	Query using GET requests. Filters are passed as GET parameters in the URL.	✓